Beyond Textualization: An Oracle Study of Evidence Representation in Multimodal Document Question Answering

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Abstract

We study how evidence representation alone affects answer quality in multimodal document question answering. Retrieval is held fixed (gold qrels, top-1) so the comparison isolates how a vision-language decoder uses different evidence sources at the answer stage. With Qwen2-VL-7B-Instruct as the generator under NF4 4-bit quantisation, we compare seven evidence modes on four VisRAG-Ret-Test public splits (InfoVQA, ChartQA, MP-DocVQA, SlideVQA): image-only, text-only with self-OCR or Upstage Document Parse, selective hybrid variants that route chart pages to image evidence, and image+text combinations on both text sources. We score relaxed Exact Match and a Qwen2.5-VL-72B-judged faithfulness score. Image-grounded modes dominate textualisation on both metrics, and Image+Text with self-OCR is the only mode that improves macro EM and macro faithfulness over Image-only (+2.53 EM, +0.44 faithfulness percentage points). Selective routing helps when in-model OCR is noisy but does not help structured parser output. The reference comparison shows an ordering inversion: with text alone, Upstage Document Parse beats the model's self-OCR by +14.7 macro EM percentage points, while in the reference image+text condition self-OCR is ahead of Upstage Document Parse — by +10.7 macro EM percentage points in the team's tf457 numbers and by +1.85 macro EM percentage points on the clean transformers 4.57.6 re-run reported in Section 5. A paired follow-up diagnostic probes why Image+Text with Upstage Document Parse underperforms its self-OCR counterpart: an image-first prompt swap and a shallow surface-form normalisation both fail to close the gap (McNemar p=0.11 and p=1.00), constraining the explanatory space for follow-up work.

1. Introduction

Multimodal document question answering combines optical character recognition, layout understanding, and visual reasoning over rendered page images. A modern recipe such as VisRAG [1] retrieves page images and feeds them to a vision-language generator end-to-end, skipping the explicit OCR step that token-level pipelines such as LayoutLMv3 [10] depend on. End-to-end evaluations of such systems mix two effects: how well the retriever finds the right page, and how well the generator uses the page once it has it. The two are easy to confound on small benchmarks, and the latter is what a system designer can act on if the retriever is already strong.

We isolate the second effect with an oracle-retrieval study. We fix retrieval to gold qrels at top-1 across all experiments, so the correct page is always served and only the answer stage is exercised. Within this controlled setting we ask four research questions that span the evidence-representation design space. RQ1: Which context modality is most reliable when the model already perceives the page? RQ2: Can OCR text replace image evidence in either direction (in-model self-OCR or a production parser)? RQ3: Does selectively filtering text evidence reduce OCR noise without losing information? RQ4: Does evidence representation affect unsupported hallucination, not just exact match? RQ4 motivates pairing relaxed EM with a faithfulness score judged by a stronger vision-language model.

We compare seven evidence modes on four VisRAG-Ret-Test public splits (InfoVQA, ChartQA, MP-DocVQA, SlideVQA) using Qwen2-VL-7B-Instruct [6] as the generator under NF4 4-bit quantisation [11]. The two text sources we consider are the generator's own self-OCR transcription and the Upstage Document Parse [13] commercial parser. Image-grounded modes dominate textualisation on both EM and faithfulness; Image+Text with self-OCR is the only mode that improves macro EM and macro faithfulness over Image-only. After running the seven-way comparison we noted that Image+Text with Upstage Document Parse underperforms its self-OCR counterpart on the same image — by +10.7 macro EM percentage points in the team's tf457 numbers. The same cell re-run on a clean transformers 4.57.6 environment gives a smaller +1.85 macro EM percentage-point gap, so the original figure is partly an environment artefact (Section 5); the direction of the comparison is preserved across both environments. A short follow-up diagnostic probes this underperformance with a paired prompt swap and a shallow surface-form normalisation.

Contributions. The contributions of this paper are as follows:

C1. An oracle-controlled empirical study isolating evidence representation from retrieval failure in multimodal document QA. The study uses gold qrels at top-1 across four VisRAG-Ret-Test public splits with Qwen2-VL-7B-Instruct.

C2. A seven-way comparison across four evidence representations (image-only, text-only, selective hybrid, image+text) and two text sources (in-model self-OCR and a production document parser), evaluated on both EM and a Qwen2.5-VL-72B-judged faithfulness score. Image-grounded modes dominate textualisation on both metrics; Image+Text with self-OCR is the only mode that improves macro EM and macro faithfulness over Image-only.

C3. Evidence that selective routing of OCR text helps when in-model OCR is noisy (chart pages on Qwen self-OCR) but does not help structured parser output (Upstage Document Parse). This implies a routing strategy should depend on the text source, not on the page type alone.

C4. A paired follow-up diagnostic on the underperforming Image+Text (Upstage Document Parse) cell. A prompt swap that demotes text and elevates the image and a shallow surface-form normalisation of the parser output both fail to close the gap at $\alpha=0.05$ (McNemar $p=0.11$ and $p=1.00$ respectively), constraining the explanatory space and motivating three candidate hypotheses for follow-up work.

2. Related Work

VisRAG [1] proposes vision-based retrieval and image-only generation for document QA, with code and benchmark data released through the OpenBMB/VisRAG repository [16]. Its public retrieval-test splits are built on DocVQA [2], InfographicVQA [3], ChartQA [4] and SlideVQA [5]; we use the four splits as our oracle evaluation harness. A complementary line of work flattens the page into text and layout tokens: LayoutLMv3 [10] pre-trains a transformer on OCR tokens with masked-image modelling, and commercial systems such as Upstage Document Parse [13] expose a production parser with explicit element types and bounding boxes.

Recent vision-language models such as Qwen2-VL [6], Qwen2.5-VL [7], InternVL 2.5 [12] and LLaVA [8] now serve as joint readers and generators over rendered document pages, which raises the question studied here: when the generator can already see the page, what does an external transcription add? Earlier work tended to assume the addition was either neutral (a textual transcription stabilises named entities and numbers) or strictly helpful (more evidence is better than less). Our seven-way comparison in Section 4 shows that neither assumption holds uniformly: textualisation strictly underperforms the page image at our scale, and adding parser text to the image can even reduce both EM and faithfulness relative to the image-only baseline.

Evidence fusion has been studied in the knowledge-VQA setting [9], where retrieved text passages are mixed with a visual query, and in the broader RAG literature for text-only tasks. Recent work in long-context language modelling reports that the position and length of supporting text affect answer accuracy even when the answer is present somewhere in the context [14], an observation that informs the length-as-distractor hypothesis we sketch in Section 6. Prompt sensitivity of vision-language models is itself an active area; the order in which evidence is presented has been reported to shift answers, and chain-of-thought prompting [15] is a related intervention. These observations motivate the paired prompt-bias and surface-form diagnostics in Section 5. Finally, while a number of multimodal QA papers report end-to-end scores, comparatively few isolate the answer stage by holding retrieval fixed at gold top-1. Doing so is the methodological backbone of the present study and lets us attribute every cross-mode gap to the answer-stage evidence representation rather than to retrieval noise.

3. Method and Experimental Setup

We frame evaluation as an oracle-retrieval study. For every query we serve the gold page from the dataset's qrels and ask the generator to produce a short answer. This removes retrieval as a confounder so the comparison between evidence modes isolates the answer stage. Algorithm 1 sketches the procedure: for each (query, gold-page) pair and each evidence mode m with text source s, we assemble the evidence content $E^{(m,s)}$ and wrap it with the mode-specific instruction clause ι_m to form the prompt, then decode a 20-token answer.

Algorithm 1 (oracle evidence-mode evaluation). Let $\mathcal{D}$ denote the set of evaluation splits and $d \in \mathcal{D}$ an individual split with n_d queries. For each $d \in \mathcal{D}$ and each query $q \in d[:n_d]$: (1) look up the gold page $p^* = \mathrm{qrels}(q)$ and render $I = \text{page-image}(p^*)$; (2) for each evidence mode m and text source $s \in \{\text{self}, \text{upstage}\}$, build the prompt $P^{(m,s)}(q, I)$ by instantiating the shared skeleton with evidence E^(m,s)(q, I, T^(s)) and the instruction clause ι_m; (3) decode answer â^(m,s) = f_θ(P^(m,s)(q, I)) where f_θ is Qwen2-VL-7B, with do_sample=False, max_new_tokens=20; (4) score relaxed EM against the gold answer and, where applicable, score faithfulness with a Qwen2.5-VL-72B judge; (5) aggregate per-mode macro accuracy across $\mathcal{D}$.

$$p_i^* = \mathrm{qrels}(q_i), \qquad I_i = \mathrm{render}(p_i^*) \tag{1}$$

3.1 Evidence modes

We compare seven evidence modes that bracket the design space: (i) Image-only: only the gold page image is supplied. (ii) Text-only (Qwen self-OCR): Qwen2-VL-7B is run on the page in an OCR-extraction pass with a transcription prompt; only the resulting text is supplied. (iii) Text-only (Upstage Document Parse): the page is parsed by the Upstage Document Parse [13] service and only the parser text is supplied. (iv) Selective Hybrid (Qwen self-OCR): an LLM router classifies each page as text/mixed/chart, supplies Qwen self-OCR text on text and mixed pages, and routes chart pages to the page image. (v) Selective Hybrid (Upstage Document Parse): the same router supplies Upstage text on text and mixed pages and routes chart pages to the page image. (vi) Image+Text (Qwen self-OCR): the page image and the Qwen self-OCR transcription are supplied together. (vii) Image+Text (Upstage Document Parse): the page image and the Upstage Document Parse output are supplied together. All seven modes share the same answer-generation prompt skeleton; only the evidence-instruction clause changes. Figure 1 sketches the oracle-controlled evidence pipeline that produces E for each mode.

$$T_i^{(s)} = g_s(I_i), \qquad E_i^{(m,s)} = \varphi_m(q_i, I_i, T_i^{(s)}), \quad s \in \{\mathrm{self}, \mathrm{upstage}\} \tag{2}$$

$$E_i^{(m,s)} = \begin{cases} \{I_i\}, & \mathrm{m} = \mathrm{image-only} \\ \{T_i^{(s)}\}, & \mathrm{m} = \mathrm{text-only} \\ \{T_i^{(s)}\}, & \mathrm{m} = \mathrm{sel.}, r_i \neq \mathrm{chart} \\ \{I_i\}, & \mathrm{m} = \mathrm{sel.}, r_i = \mathrm{chart} \\ \{I_i, T_i^{(s)}\}, & \mathrm{m} = \mathrm{image + text} \end{cases} \tag{3}$$

Here s ∈ {self, upstage} selects the text source: g_self is Qwen2-VL-7B in an OCR pass (the in-model self-OCR), and g_upstage is the Upstage Document Parse service. The image-only mode does not consume a text source, so s is unused there. The router label r_i = r(q_i, I_i) is used only by the selective hybrid modes; non-chart pages use text evidence under the relevant source s, while chart pages fall back to the page image.

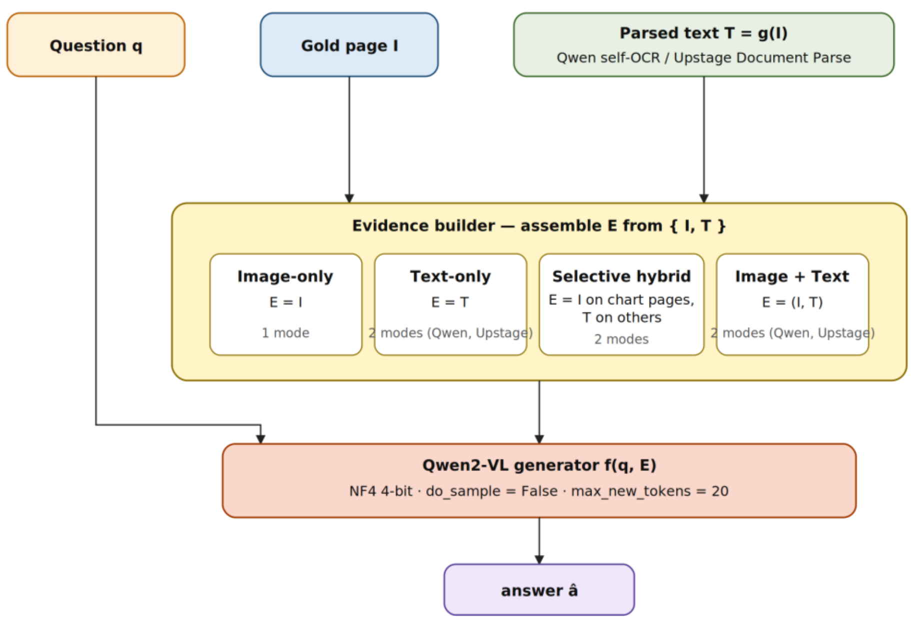

Figure 1. Oracle-controlled evidence pipeline. Question q is paired with the gold page I (oracle qrels, top-1) and the parsed text T = g(I), where g is either Qwen self-OCR or Upstage Document Parse. The evidence builder assembles the evidence set E according to the selected mode; the Qwen2-VL generator decodes the answer from (q, E) under NF4 4-bit quantisation with do_sample=False and max_new_tokens=20. Only the contents of E vary across the seven modes.

3.2 Answer-generation prompts

All seven modes share a fixed answer-generation skeleton that instructs the model to produce a short, factual answer; only the evidence-instruction clause varies across modality families (image-only, text-only, image+text). For Image-only, the model is asked to answer directly from the page image. For Text-only, it answers from parsed text/table evidence. For Image+Text, parsed text is framed as support for exact labels and values while the image remains available for visual verification. The OCR-extraction and faithfulness-judge prompts are unchanged across conditions; verbatim templates are provided in the supplementary archive.

$$
\begin{aligned}
P_i^{(m)} &= \mathrm{prompt}(q_i, E_i^{(m)}, \iota_m), \\
\hat{a}_i^{(m)} &= f_\theta(P_i^{(m)}).
\end{aligned}
\tag{4}
$$

3.3 Datasets

We evaluate on four VisRAG-Ret-Test public splits: InfoVQA [3], ChartQA [4], MP-DocVQA (DocVQA [2] multi-page), and SlideVQA [5]. We use the first 100 queries of each split, except ChartQA, which has 63 queries due to qrels coverage in the public release of VisRAG-Ret-Test-ChartQA. The total evaluation set therefore contains 363 queries (100 + 63 + 100 + 100).

3.4 Backbone, quantisation, and decoding

All seven evidence modes use Qwen2-VL-7B-Instruct [6] loaded with bitsandbytes NF4 4-bit quantisation [11]. Decoding uses do_sample=False, max_new_tokens=20, and max_pixels=1280×1024 for image inputs. The faithfulness judge is Qwen2.5-VL-72B [7]. Faithfulness is not re-scored for the diagnostic cells in Section 5 because the 72B judge model is expensive to host; the diagnostic comparisons are reported on EM only.

Environment note. The reference seven-way run reported in Table 1 was produced under transformers 4.57.6, except for the Image+Text (Upstage Document Parse) cell, which was carried over from an earlier transformers 4.51 run that contained a known Qwen2-VL vision-encoder bug (cf. the Table 1 caption and the post-Table-1 paragraph in Section 4). All diagnostic re-runs in Section 5.1 (prompt-bias) and Section 5.2 (surface-form normalisation) use transformers 4.57.6 end-to-end. Whenever Section 5 reports a macro EM that differs from Table 1, the difference is attributable to this environment correction, not to a change in samples, prompts, or the backbone.

3.5 Selective routing details

The selective hybrid modes (iv) and (v) use a single LLM router that classifies each gold page as text, mixed, or chart from a low-resolution view of the page image. Text and mixed pages are served as text-only evidence under the relevant text source (Qwen self-OCR or Upstage Document Parse). Chart pages are routed to image evidence — the page image itself is supplied to the generator in place of a text transcription — because in-model OCR on a chart is the worst case for textualisation. The router is held fixed across the two text sources so that any difference between modes (iv) and (v) is attributable to the text source rather than to routing decisions.

3.6 Metrics

We report relaxed Exact Match (EM) and a faithfulness score. Relaxed EM matches our implementation in benchmark/metrics.py: predictions and the stringified gold answer are lowercased and whitespace-normalised; numerical tokens are matched within a 5% relative tolerance; otherwise substring containment in either direction is required. A list-style gold answer is compared by substring containment against its rendered list. The faithfulness score is the average of a per-query {0, 0.5, 1} judgement from the Qwen2.5-VL-72B judge, where 1 means the predicted answer is fully supported by the evidence, 0.5 means partially supported, and 0 means unsupported or contradicted. Macro accuracy is the unweighted mean of per-dataset scores.

$$
S_{i,d}^{(m,s)} = \begin{cases} \mathrm{EM}(\hat{a}_{i,d}^{(m,s)}, a_{i,d}), & S = \mathrm{EM} \\ \mathrm{J}(q_{i,d}, E_{i,d}^{(m,s)}, \hat{a}_{i,d}^{(m,s)}), & S = \mathrm{Faith} \end{cases}
\tag{5}
$$

$$
\mathrm{Macro}_S(m,s) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{1}{n_d} \sum_{i=1}^{n_d} S_{i,d}^{(m,s)},
\tag{6}
$$

$$
\Delta_S(m,s) = 100\,[\,\mathrm{Macro}_S(m,s) - \mathrm{Macro}_S(\text{image} - \text{only})\,].
\tag{7}
$$

Here S denotes either relaxed EM (a string-based score against the gold answer) or the faithfulness judge J that takes the query, the supplied evidence and the predicted answer and returns a {0, 0.5, 1} support score. Δ_S(m, s) in Eq. (7) is the percentage-point difference of

Macro_S(m, s) against the Image-only baseline, which has no text source and is therefore indexed by m only. Δ_S(m, s) is reported in the final column of Table 1.

3.7 Statistical analysis

For the follow-up diagnostics in Section 5 we report 95% bootstrap confidence intervals around macro accuracy (2000 iterations, per-dataset stratified resample, percentile method) and paired two-sided McNemar tests with continuity correction over the n=363 union of the four dataset subsets. The bootstrap and McNemar computations are implemented in benchmark/stats_prompt_test.py.

4. Results

Table 1 reports relaxed EM and faithfulness for all seven evidence modes on the four splits, together with the macro Δ vs the Image-only baseline for both metrics. Figure 2 visualises the text-only vs image+text ordering between the two text sources so that the inversion behind RQ1 is immediately visible. We then go through the four research questions in turn.

Table 1. Seven-way evidence-mode comparison. Cells: relaxed EM / Faith. Last column: macro Δ vs Image-only (pp). † Carried over from a transformers 4.51 run with a Qwen2-VL vision-encoder bug; clean 4.57.6 re-run reported in Table 2.

Method	Info	Chart	MP-Doc	Slide	Avg Δ
Image	0.720 / 0.780	0.556 / 0.730	0.880 / 0.900	0.540 / 0.850	— / —
Text (Qwen)	0.340 / 0.360	0.318 / 0.222	0.670 / 0.790	0.430 / 0.545	−23.45 / −33.58
Text (Upstage)	0.540 / 0.620	0.524 / 0.587	0.760 / 0.939	0.520 / 0.695	−8.79 / −10.46
Sel. Hybrid (Qwen)	0.340 / 0.350	0.524 / 0.540	0.670 / 0.780	0.440 / 0.565	−18.04 / −25.64
Sel. Hybrid (Upstage)	0.540 / 0.630	0.460 / 0.619	0.770 / 0.900	0.480 / 0.680	−11.13 / −10.78
Image+Text (Qwen)	0.730 / 0.770	0.667 / 0.778	0.820 / 0.880	0.580 / 0.850	+2.53 / +0.44
Image+Text (Upstage) †	0.540 / 0.650	0.508 / 0.524	0.800 / 0.900	0.520 / 0.730	−8.19 / −11.41

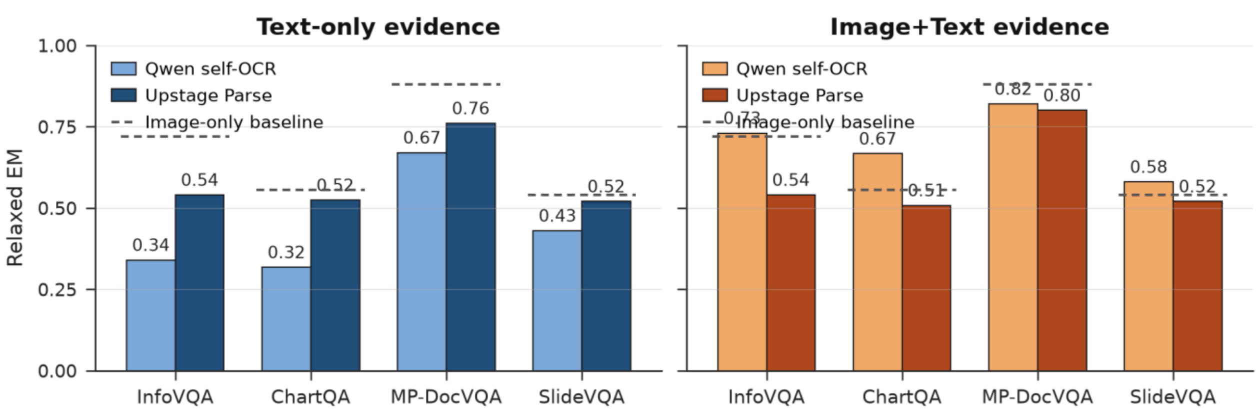

Figure 2. Relaxed EM per dataset by text source and evidence mode.

The Image+Text (Upstage Document Parse) row in Table 1 reads a macro EM Δ of −8.19 percentage points against Image-only. Section 5 reports a clean re-run of this cell on transformers 4.57.6 with macro EM 0.7061, which is +3.22 percentage points above the Image-only baseline rather than −8.19. The direction of the cross-source comparison (Image+Text self-OCR ≥ Image+Text Upstage Document Parse) is preserved between the two environments, but the magnitude of both the cross-source gap and the Image-only comparison shrinks substantially once the 4.51 vision-encoder bug is removed. We retain the original team-reported numbers in Table 1 because they are the published reference, and use the clean re-run in Tables 2–3 as the basis for the diagnostic statistical claims.

4.1 RQ1: Which Context Modality Is Most Reliable?

The best context is visually grounded. Image+Text with Qwen self-OCR reaches the best macro EM and the best macro faithfulness in Table 1, improving on Image-only by +2.53 EM and +0.44 faithfulness percentage points, while Image+Text with Upstage Document Parse does not exceed Image-only on macro EM. Adding text helps only when it supports, rather than competes with, visual grounding; Section 5 returns to the underperforming Image+Text (Upstage Document Parse) cell.

4.2 RQ2: Can OCR Text Replace Image Evidence?

OCR text alone cannot replace image evidence. Text-only with Qwen self-OCR loses 23.45 macro EM and 33.58 macro faithfulness against Image-only. Upstage Document Parse is the better text source on its own (+14.7 macro EM over Qwen self-OCR text-only), but even Upstage text-only sits 8.79 macro EM and 10.46 macro faithfulness percentage points below Image-only. Replacing the page image with text is a strict downgrade at Qwen2-VL-7B scale on this benchmark family.

4.3 RQ3: Does Selective Hybrid Routing Reduce OCR Noise?

Selective routing helps when OCR is noisy but is not universally reliable. Selective routing of Qwen self-OCR improves ChartQA EM from 0.3175 to 0.5238 (a +0.21 absolute jump) because routing chart pages out of the text channel avoids a noisy self-OCR pass. Selective routing of Upstage Document Parse does not produce consistent gains across splits: parser output is already structured, and dropping pages can remove useful evidence on a split-by-split basis. The headline implication is that a routing strategy should depend on the text source, not only on the page type. If the text source is in-model OCR, routing chart pages out of the OCR pipeline pays off; if the text source is a structured parser, routing tends to drop information without a clean gain.

The asymmetry is diagnostic: in-model OCR loses spatial structure on chart pages, so filtering them out removes the worst failure mode, whereas parser output preserves textual labels and table-like cells, so dropping chart pages sacrifices structured information without an OCR-noise benefit. The takeaway is that page-type routing decisions cannot be re-used across text sources.

4.4 RQ4: Does Evidence Representation Affect Unsupported Hallucination?

Evidence representation drives unsupported hallucination, not just accuracy. Text-only with Qwen self-OCR has the lowest faithfulness (macro Δ −33.58). Image+Text with Qwen self-OCR is the only mode that improves macro faithfulness over Image-only (+0.44), while Image+Text with Upstage Document Parse shows the opposite pattern (macro Δ −11.41): the long, layout-preserving parser output appears to distract from image-based verification. This cell motivates Section 5.

5. Follow-up Diagnostics on the Image+Text (Upstage Document Parse) Cell

Section 4 noted that Image+Text with Upstage Document Parse underperforms Image+Text with Qwen self-OCR on the same image, even though Upstage Document Parse is the better text source on its own. We probe two candidate causes of that underperformance with paired diagnostics. Both diagnostics use the same 100-query splits (63 for ChartQA) and the same backbone and decoding configuration as Section 4. Faithfulness is not re-scored for the diagnostic cells because the 72B judge model is expensive to host; the diagnostic comparisons are reported on EM only.

5.1 Diagnostic 1: Prompt-Bias Intervention

Table 2 reports our clean transformers 4.57.6 re-run of the four Image+Text cells. The Upstage orig. row in Table 2 reads macro EM 0.7061 against Table 1's 0.5920 for the same cell; the difference is the transformers 4.51 vision-encoder bug that the older run carried, not a change in method, sample, or prompt. The Image+Text instruction clause in Section 3.2 asks the model to use parsed text for exact labels and the image as a visual check. If the parsed text is clean — which it is for Upstage Document Parse — the decoder can satisfy the prompt without ever attending to the image. The diagnostic swaps this clause for an image-first instruction:

> *“Use the document page image(s) as the primary evidence. Read directly from the image. Use the parsed text/table only as a secondary cross-check for ambiguous labels or numeric values.”*

The image-first prompt was applied to both text sources on all four splits, giving a 4 × 2 × 2 = 16-cell sweep (datasets × text source × prompt). Each cell is the first 100 queries (63 for ChartQA), oracle top-1, do_sample=False, max_new_tokens=20. Macro accuracy is reported with 95% bootstrap CIs. Results are in Table 2.

Table 2. Diagnostic 1: image-first prompt swap. EM with 95% bootstrap CIs.

Source	Prompt	InfoVQA	ChartQA	MP-DocVQA	SlideVQA	Macro [95% CI]
Upstage	orig.	0.680	0.714	0.860	0.570	0.706 [0.660, 0.753]
Upstage	image-first	0.690	0.714	0.880	0.600	0.721 [0.675, 0.767]
Qwen	orig.	0.710	0.698	0.860	0.630	0.725 [0.679, 0.772]
Qwen	image-first	0.730	0.698	0.890	0.620	0.735 [0.690, 0.780]

These paired diagnostic rows show the same direction as the main table (Upstage Document Parse trails Qwen self-OCR by 1.85 macro percentage points here vs. 10.72 in Table 1), but at smaller magnitude. The residual gap is statistically fragile: a paired McNemar test (continuity-corrected, two-sided) on n=363 queries gives p=0.20 for Upstage Document Parse vs Qwen self-OCR under the original prompt, and p=0.36 under the image-first prompt. The prompt-swap test (image-first vs original within the same evidence source) gives p=0.11 for Upstage Document Parse and p=0.39 for Qwen self-OCR. None reach α=0.05.

5.2 Diagnostic 2: Shallow Surface-Form Normalisation

Qwen self-OCR text is sampled from the same decoder that will later answer the question, so its surface form (tokens, spacing, line-wrap conventions) matches what the decoder expects to see. Upstage Document Parse output is layout-preserving prose with bullet markers, section labels, dash rules, and visual line-wraps that the Qwen decoder would not itself produce. The hypothesis is that this surface mismatch costs cross-modal alignment in the Image+Text setting. We applied a deliberately shallow normaliser to the Upstage text before it entered the prompt: strip parser markers (bullet glyphs such as ●, a Korean section tag the parser inserts for collected content, and horizontal dash rules separating layout regions), collapse intra-line whitespace, rejoin visual line-wraps. Length deltas after normalisation are tiny: InfoVQA 18,304→18,244 characters (−0.3%), ChartQA 1,405→1,405 (0%), MP-DocVQA 2,693→2,691 (−0.07%), SlideVQA 2,217→2,213 (−0.18%).

Table 3. Diagnostic 2: surface-form normalisation on Upstage Image+Text. EM with 95% bootstrap CIs.

Method	InfoVQA	ChartQA	MP-DocVQA	SlideVQA	Macro [95% CI]
Original	0.680	0.714	0.860	0.570	0.706 [0.660, 0.753]
Normalised	0.680	0.714	0.860	0.580	0.709 [0.661, 0.754]

The paired test yields only three discordant pairs (1 baseline-only correct, 2 normalised-only correct), giving p=1.00 under McNemar with continuity correction. The 95% bootstrap CIs of the two rows overlap heavily. The intervention is, statistically, a null.

5.3 Combined Verdict

Neither cheap intervention closes the Image+Text (Upstage Document Parse) gap, and neither settles the deeper mechanism. Section 6 lists three co-equal candidate hypotheses, each with a follow-up experiment that could falsify it.

6. Discussion

Re-stating contributions in light of the results. C1: the oracle-controlled setup successfully isolates evidence representation from retrieval; the cross-mode gaps in Table 1 are large enough to be informative on 363 queries. C2: the seven-way comparison shows that image-grounded modes dominate textualisation on both EM and faithfulness, with Image+Text (Qwen self-OCR) the only mode that improves both macro metrics over Image-only. C3: selective routing is text-source-dependent; it helps in-model OCR on chart pages and does not consistently help structured parser output. C4: the paired diagnostics on the Image+Text (Upstage Document Parse) cell do not support prompt-driven evidence reweighting or shallow surface alignment as sufficient explanations, but they do not pin down a single mechanism.

Three candidate hypotheses for the residual gap remain, presented as co-equal. (i) Representational misalignment: self-OCR text is close to the decoder's own distribution, while Upstage Document Parse output comes from a different model and may be off-distribution for cross-modal attention. Falsification: rewrite the parser text through Qwen itself before serving it; a closed gap supports alignment-to-self-distribution. (ii) Modality attention competition: clean external text saturates the answer-grounding pathway and the image gets under-attended. Falsification: log cross-modal attention weights under the two text sources on the discordant queries. (iii)

Length as distractor: on InfoVQA the parser text is an order of magnitude longer than on the other three splits (18 k characters vs 1.4 k–2.7 k). Falsification: a length-controlled re-run on InfoVQA, truncating the parser text to the length of the self-OCR text.

7. Limitations

Several limitations narrow the claim. (i) A single generator family (Qwen2-VL-7B-Instruct) under NF4 4-bit quantisation. Our findings should not be read as general claims about vision-language models. The ordering and the residual gap may behave differently for larger Qwen2-VL/2.5-VL variants, for non-Qwen families such as InternVL or LLaVA, or for non-quantised inference. We mark this as the single largest threat to the external validity of the paper. (ii) A single seed; do_sample=False but CUDA non-determinism can produce a few percentage points of run-to-run drift. (iii) 100 samples per split (63 for ChartQA) for a total n=363, which is the source of the wide bootstrap CIs in Section 5. (iv) Faithfulness was not re-scored for the diagnostic cells in Section 5 due to the cost of hosting the 72B judge model; we report EM only for those cells. A judge-aware diagnostic re-run is straightforward to add when judge compute is available. (v) The surface-form normaliser in Section 5.2 is deliberately shallow, so the null result speaks only to the easy-to-fix part of the surface-alignment hypothesis. (vi) We use oracle qrels with top-k=1, so retrieval is not exercised; end-to-end evaluation with a realistic retriever is left to future work.

8. Conclusion

We report an oracle-controlled study of evidence representation for multimodal document question answering. Across four VisRAG-Ret-Test public splits and seven evidence modes, image-grounded modes dominate textualisation on both relaxed EM and a Qwen2.5-VL-72B-judged faithfulness score; Image+Text with Qwen self-OCR is the only mode that improves macro EM and macro faithfulness over Image-only. Selective routing of OCR text helps when in-model OCR is noisy but not when the text source is a structured parser. A paired follow-up diagnostic on the underperforming Image+Text (Upstage Document Parse) cell does not support prompt-bias swap or shallow surface-form normalisation as sufficient explanations (McNemar p=0.11 and p=1.00 respectively), leaving representational misalignment, modality attention competition, and length-as-distractor as co-equal candidate hypotheses for follow-up work. Per-query predictions, the diagnostic scripts, and the runner configuration are made available in the anonymous supplementary archive that accompanies this submission (not an external URL).

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